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Localized Residual Structure in SPARC Rotation Curves: A Burkert-Backbone Model with BIC-Selected Shells

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ABSTRACT

Galactic dark matter halos are conventionally modeled with smooth profiles (NFW, Burkert, pseudo-isothermal variants), but rotation curves frequently display localized features — inner cusps, mid-disk inflections, and outer-flattening structures — that smooth profiles often struggle to reproduce without introducing global distortions or large residuals. We hypothesize that some rotation-curve features may reflect localized deviations from smooth halo structure, and that capturing them requires architectural decomposition rather than additional profile flexibility. We implement a framework comprising a Burkert backbone plus 0–2 BIC-selected Gaussian shells per galaxy, with shell width strictly constrained to $\sigma/r_{\rm shell} \leq 0.4$ via reparameterization, and test it against 102 SPARC galaxies (T=2–9, 2334 data points) spanning Sab through Sdm spirals. The framework adequately fits 91/102 galaxies ($\chi^2_{\rm red} < 1.5$) versus 52/102 for NFW and 65/102 for Burkert alone. By Bayesian Information Criterion, the framework is very strongly preferred (ΔBIC < -10) over NFW in 53 of 102 galaxies (median ΔBIC $= -35$); 8 galaxies BIC-prefer NFW, none very strongly. Comparison against Burkert alone is structurally asymmetric (the framework reduces exactly to Burkert when zero shells are selected) and we therefore report it for completeness rather than as an independent test. Shell-bearing fraction shows a statistically significant declin-

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ing trend across the morphological range (T-label permutation test on 102 individual galaxies, 10,000 resamples: $p_{\text{one-sided}} = 0.001$, $p_{\text{two-sided}} = 0.002$; Spearman $\rho = -0.83$ with bootstrap 95% CI $[-0.95, -0.22]$; Mann-Kendall $p = 0.003$): 100% at T=2 (Sab), 53–56% across T=3–6, 36% at T=7 (Scd), 33% at T=8 (Sd), and 31% at T=9 (Sdm). The signal is driven by the T=2 excess and the late-type decline; the middle of the gradient (T=3–6) is essentially flat. A synthetic null test on a stratified subsample across T=2–9 (346 mocks total) yields aggregate false shell-selection rates of 4.0% under Burkert truth and 63.6% under NFW truth, indicating that BIC suppresses shells well when the backbone matches but can misidentify base-profile shape mismatches as localized structure. The Burkert-truth false-positive rate is uniformly low ($\leq 5\%$ in 6 of 8 T-bins, with isolated rates of 11% at T=8 and 13% at T=9 reflecting low-S/N late-type rotation curves) and uncorrelated with T-type (Spearman $\rho = +0.33$, $p = 0.42$). Real-data shell-bearing fractions of 31–100% per T-bin exceed the corresponding Burkert-truth FP rates by factors of 6–20, ruling out smooth-Burkert truth as the dominant driver of the gradient. The NFW-truth null FP rate is roughly flat across T-type and could not produce the strong negative real trend either. Mixed-truth and external-family scenarios (smooth profiles outside the cored-cuspy family) remain logically possible and are addressed only partially, by the Einasto-backbone robustness test described below. The T=2 shell-bearing fraction is overwhelmingly significant against null expectations (Fisher one-sided $p \approx 10^{-6}$). At the late-type end, T=9 real shell-bearing (31%) sits significantly below NFW-truth null expectation (67%, Fisher one-sided $p = 0.05$). NGC 5055 provides a clean single-galaxy example of where smooth halos fail: Burkert $\chi^2_{\text{red}} = 9.5$, NFW $\chi^2_{\text{red}} = 30.2$, DC14 $\chi^2_{\text{red}} = 9.0$, while the framework with one BIC-selected shell at $r \approx 12$ kpc, $M \approx 4 \times 10^{10}$ $M_\odot$ achieves $\chi^2_{\text{red}} = 1.5$. As an additional test of smooth-profile expressive capacity, we fit the cosmologically-motivated Di Cintio et al. (2014a,b) DC14 feedback-modified profile to all 10 universal-failure galaxies; DC14 also fails adequacy on 10/10, with 8/10 fits pegged at the DC14 parameter boundary

attempting to revert to NFW. We note that the universal-failure subset is dominated by massive spirals (typically $V_{\rm flat} > 200$ km/s) sitting near the lower edge of the regime in which DC14 was calibrated, so this result does not generalize to the full range of feedback-modified profiles. As a backbone-robustness check (§3.7), we re-ran the framework on the full 102-galaxy sample with the Burkert backbone replaced by the more flexible 3-parameter Einasto profile; shell-bearing classification agrees with the Burkert-backbone result in 90 of 102 galaxies (88%), the morphology gradient is preserved at full-sample statistical strength (per-galaxy Spearman $\rho = -0.347$, $p = 0.0004$ under Einasto, vs $\rho = -0.296$, $p = 0.003$ under Burkert), and NGC 5055 still requires shells under the Einasto backbone (Einasto-only $\chi^2_{\rm red} = 8.42$, Einasto+2-shell $\chi^2_{\rm red} = 0.51$), indicating that the framework's main findings are not driven by the specific choice of Burkert backbone within the cored-cuspy family. Per-galaxy event identification cannot be supported by rotation curve data alone; the framework's empirical content lies in its population-level patterns and the architectural shape predictions smooth halos cannot reproduce.

Keywords: galaxies: kinematics and dynamics — dark matter — galaxies: haloes — galaxies: spiral

1. INTRODUCTION

Galactic rotation curves provide one of the clearest empirical signatures of mass discrepancies in disk galaxies. Across a wide range of galaxy masses and morphologies, observed circular velocities remain elevated beyond the radius where luminous baryons alone would predict decline. This behavior is conventionally modeled by embedding galaxies in smooth dark matter halos, with commonly used profiles including NFW (Navarro, Frenk & White 1996, 1997), Burkert (Burkert 1995), pseudo-isothermal, Einasto, and related variants. Tight relationships between baryonic and dynamical properties have also been characterized statistically across SPARC (Lelli, McGaugh & Schombert

2016; McGaugh, Lelli & Schombert 2016), providing a useful population-level reference against which residual structure can be assessed.

Smooth profiles are powerful because they capture the broad radial structure of dark matter halos with few parameters. However, many high-quality rotation curves also exhibit localized deviations from smooth behavior: inner excesses, mid-disk inflections, shoulders, and outer flattening features (for a review of rotation-curve diversity and core/cusp systematics, see de Blok 2010). These structures are often absorbed into global halo parameters, attributed to baryonic uncertainty, or treated as object-specific irregularities. Yet if such features occur systematically, then smooth profiles may be missing an architectural component of halo structure rather than merely requiring better parameter tuning.

In this work, we test the hypothesis that some rotation-curve features reflect localized residual structure that smooth halo profiles cannot capture. We do not assume that individual features can be uniquely associated with specific historical events, nor that they are necessarily of halo origin. Instead, we ask a population-level question: does allowing a small number of localized residual components improve rotation-curve fits in a statistically preferred way, and does the occurrence of such components correlate with galaxy morphology in a manner consistent with hierarchical assembly?

We construct a minimal architectural extension of a smooth halo model. The base profile is a Burkert halo, chosen because it provides a flexible cored backbone for disk-galaxy rotation curves. We then allow zero, one, or two Gaussian shell components, with shell number selected per galaxy by the Bayesian Information Criterion. The shell width is constrained by $\sigma/r \leq 0.4$ so that shells remain localized and cannot simply mimic an additional smooth halo component. When zero shells are selected, the model reduces exactly to Burkert-only. Throughout this paper we use "shell" as a generic label for these BIC-selected localized Gaussian residual components in the model. The terminology is not intended to invoke a specific physical origin (tidal-debris stellar shells, accretion remnants, or otherwise); the components are mass-density features in the dark matter inferred from rotation-curve residuals, and their physical interpretation is deferred to §5.

We apply this framework to 102 SPARC (Lelli, McGaugh & Schombert 2016) galaxies with morphological types T=2–9, spanning Sab through Sdm spirals. The comparison models are Burkert-only and free-concentration NFW, each fit on the same data subsets with the same multi-restart optimization procedure.

The central result is that the Burkert-plus-shell framework adequately fits 91/102 galaxies, compared with 65/102 for Burkert-only and 52/102 for NFW. More importantly, shell-bearing fraction shows a statistically significant declining trend with morphological type (T-label permutation test on 102 individual galaxies, $p_{\text{two-sided}} = 0.002$; Spearman $\rho = -0.83$ with bootstrap 95% CI $[-0.95, -0.22]$; Mann-Kendall $p = 0.003$), driven primarily by a statistically significant excess in early-type (T=2) galaxies and a low fraction at the latest types. This morphology gradient is the primary empirical content of the paper. We support its statistical robustness with a synthetic null test (§3.5) showing that the Burkert-truth false-positive rate is uniformly low ($\leq 5\%$ in 6 of 8 T-bins), an order of magnitude below the real-data shell-bearing fractions in every T-bin — ruling out smooth-Burkert truth as the dominant driver of the gradient, with NFW-truth ruled out by analogous reasoning, though mixed-truth and external-family scenarios remain logically possible.

We emphasize that the present framework is phenomenological. It does not prove that any specific fitted shell corresponds to a particular merger event, satellite accretion episode, or dark-sector structure, and it does not establish that the localized components are necessarily of halo origin — they could equally reflect localized baryonic features not captured by the SPARC mass-to-light decomposition, kinematic artifacts of triaxial halo or warped-disk geometry, or non-circular motions. One-dimensional rotation curves alone do not contain enough information for such per-galaxy attribution. The claim is more limited but also more robust: BIC-selected localized residual components capture rotation-curve features that smooth profiles systematically miss, and the population-level occurrence of those components varies with morphological type in a manner consistent with hierarchical-assembly expectations and inconsistent with the simplest smooth-halo null scenarios.

The framework is also agnostic with respect to the dark-matter-versus-modified-gravity debate. SPARC has been the primary observational anchor for the radial acceleration relation and MONDstyle interpretations (McGaugh, Lelli & Schombert 2016), in which rotation-curve regularities are recast as an empirical $g_{\rm obs}(g_{\rm bar})$ relation rather than the mass distribution of an unseen halo. Our analysis treats $V_{\rm DM}^2 \equiv V_{\rm obs}^2 - V_{\rm bar}^2$ as the residual signal to be modeled and does not attempt to distinguish between dark-matter and modified-gravity interpretations of that residual; either can in principle host the localized residual structure we report. Whether shell-bearing galaxies depart systematically from the radial acceleration relation, or whether shell radii correlate with the radii at which $g_{\rm obs}/g_{\rm bar}$ deviates most strongly from the relation, are questions left for future work.

2. METHODS

2.1. *Sample and data*

The SPARC database (Lelli, McGaugh & Schombert 2016) provides Spitzer Photometry & Accurate Rotation Curves for 175 disk galaxies. We restrict our analysis to morphological types T=2–9, spanning Sab to Sdm spirals, yielding a sample of 102 galaxies with 2334 total rotation curve data points. The original analysis was conducted on T=2–7 (80 galaxies); we extended to T=8–9 (an additional 22 galaxies) to test whether the morphology gradient continues into the late-type regime where merger histories are predicted to be poorest.

For each galaxy, the SPARC rotation files provide observed velocities $V_{\rm obs}(r)$ with uncertainties $\sigma_V(r)$, gas velocity contributions $V_{\rm gas}(r)$, stellar disk contributions $V_{\rm disk}(r)$, and stellar bulge contributions $V_{\rm bulge}(r)$. The squared baryonic velocity is constructed using fixed mass-to-light ratios:

$$V_{\rm bar}^2(r) = V_{\rm gas}|V_{\rm gas}| + 0.5V_{\rm disk}|V_{\rm disk}| + 0.7V_{\rm bulge}|V_{\rm bulge}| \tag{1}$$

with $\Upsilon_{\rm disk} = 0.5$ and $\Upsilon_{\rm bulge} = 0.7$ in solar units at 3.6 μm. The dark matter velocity contribution is then

$$V_{\rm DM}^2(r) = V_{\rm obs}^2(r) - V_{\rm bar}^2(r). \tag{2}$$

When $V_{\rm bar}^2(r) \geq V_{\rm obs}^2(r)$, the inferred $V_{\rm DM}^2$ is unphysical. Such points are excluded from the fit.

2.2. *Framework architecture*

We model the dark matter velocity-squared as the sum of a Burkert backbone and zero or more discrete Gaussian shells:

$$V_{\mathrm{DM}}^2(r) = V_{\mathrm{Burkert}}^2(r;\rho_0,a) + \sum_i V_{\mathrm{shell},i}^2(r;M_i,r_i,\sigma_i). \tag{3}$$

The Burkert profile is

$$\rho_{\mathrm{Burkert}}(r) = \frac{\rho_0 a^3}{(r+a)(r^2+a^2)}. \tag{4}$$

Each shell is a Gaussian mass distribution with total mass M_i, central radius r_i, and width σ_i:

$$M_{\mathrm{shell}}(r) = \tfrac{1}{2}M_i\left[\mathrm{erf}\left(\frac{r-r_i}{\sigma_i\sqrt{2}}\right) - \mathrm{erf}\left(\frac{-r_i}{\sigma_i\sqrt{2}}\right)\right] \tag{5}$$

$$V_{\mathrm{shell}}^2(r) = GM_{\mathrm{shell}}(r)/r. \tag{6}$$

We constrain shells to be physically localized through three architectural choices: (1) $\sigma_i/r_i \leq 0.4$, enforced strictly via the reparameterization $\sigma_i = f_i\, r_i$ with $f_i \in [0.01, 0.4]$ as a box-bounded fit parameter; (2) $10^6\ \mathrm{M}_\odot \leq M_i \leq 5\times10^{10}\ \mathrm{M}_\odot$; (3) $N_{\mathrm{shells}} \in \{0,1,2\}$. The radial bound on r_i is set in the multi-restart fit procedure to $r_{\mathrm{max}} \in \{3,6,12\}$ kpc, where each value is used as a separate starting condition and the resulting fits are compared by BIC. When $N_{\mathrm{shells}} = 0$, equation (3) reduces to $V_{\mathrm{DM}}^2 = V_{\mathrm{Burkert}}^2$. Because the BIC selection (§2.3) can choose zero shells, the framework reduces exactly to Burkert-only whenever shell components are not warranted by the data.

2.3. *Fitting procedure*

For each galaxy, fits are performed using non-linear least squares (`scipy.optimize.curve_fit`) minimizing

$$\chi^2 = \sum_j \left[\frac{V_{\mathrm{obs},j} - V_{\mathrm{model},j}}{\sigma_{V,j}}\right]^2 \tag{7}$$

with σ_V floored at 1 km/s. We adopt a multi-restart procedure with 8–24 distinct initial-condition vectors per fit. The `maxfev` parameter is set to 15,000–20,000 (default is 1,000), and the fit returning the lowest χ^2 is retained.

For each galaxy, we fit the framework with $N_{\rm shells} = 0, 1, 2$ and select among configurations using the Bayesian Information Criterion:

$$\mathrm{BIC}(N) = \chi^2_{\min}(N) + p(N)\ln(n) \tag{8}$$

where $p(N) = 2 + 3N$ and n is the number of data points. We follow Kass & Raftery (1995) for interpretation of ΔBIC differences in cross-model comparisons: $|\Delta\mathrm{BIC}| < 2$ inconclusive, 2–6 positive, 6–10 strong, > 10 very strong evidence.

2.4. *Comparison models*

We fit two smooth-halo models for direct comparison. **Burkert-only** uses the same parameterization as the backbone, without shells. **NFW** (Navarro, Frenk & White 1996, 1997) uses $\rho_{\rm NFW}(r) = \rho_s/[(r/r_s)(1+r/r_s)^2]$ with free ρ_s and r_s. We do not fix concentration via cosmologically-motivated mass-concentration scalings (e.g., Dutton & Macciò 2014), because removing this degree of freedom disadvantages NFW at small samples.

2.5. *Convergence verification*

Single-restart procedures of the kind often used in nonlinear profile fitting (with `maxfev` defaults of 1,000–2,500) systematically converged to suboptimal local minima for galaxies requiring shell components. Under single-restart, NGC 2403 yielded a framework BIC of 728, while multi-restart reduced it to 193. NGC 2841 yielded NFW BIC of 1547 under single-restart, reduced to 160 under multi-restart. All fits reported here use the multi-restart procedure.

3. RESULTS

3.1. *Aggregate fit quality across T=2–9*

Across 102 galaxies and 2334 data points, the framework reduces total χ^2 by 80.0% versus Burkert-only and 76.2% versus NFW (Table 2). A first property worth flagging before quoting adequacy counts: BIC selects zero shells in 50 of 102 galaxies. In nearly half the sample, the framework reduces exactly to Burkert-only — the BIC parameter penalty effectively suppresses shells when the data does not warrant them, and the framework is not biased toward over-fitting localized structure.

The shell-bearing classifications discussed below therefore reflect genuine model preference, not an architectural tendency to add shells regardless.

We adopt $\chi^2_{\rm red} < 1.5$ as a threshold for adequate fit. This threshold is intentionally permissive relative to the formal unit-variance expectation: SPARC velocity uncertainties are known to be modestly underestimated, particularly at outer radii where beam smearing and asymmetric drift corrections introduce systematic uncertainty not captured in the quoted σ_V (Lelli, McGaugh & Schombert 2016, §3.2); $\chi^2_{\rm red} \approx 1$–1.5 corresponds in practice to fits that lie within the visible scatter of the data points. We use the same threshold for all four models so that the comparison remains internally consistent. Because adequacy counts are inevitably threshold-dependent, Table 2 also reports the count at the stricter $\chi^2_{\rm red} < 1.0$ threshold. The framework's dominance is threshold-independent within this range: at the stricter cut, 86/102 framework, 57/102 Burkert, 41/102 NFW; at the headline $\chi^2_{\rm red} < 1.5$ cut, 91/102 framework (89.2%), 65/102 Burkert-only (63.7%), and 52/102 NFW (51.0%). The gap between the framework and the smooth alternatives actually widens slightly under stricter scoring.

The framework rescues 39 galaxies that NFW fails and 26 galaxies that Burkert fails. The reverse — galaxies where NFW or Burkert adequately fit but the framework fails — occurs in zero galaxies, the latter as a consequence of the framework's superset architecture. At the late-type extreme (T=9), Burkert clearly outperforms NFW (median NFW/Burkert χ^2 ratio ≈ 4.6, Burkert wins per-galaxy in 14 of 16 cases); base-profile coupling concerns are minimal at T=9.

3.2. *Win/loss analysis*

By ΔBIC, the framework is very strongly preferred ($\Delta\mathrm{BIC} < -10$) over NFW in 53 galaxies and over Burkert in 38 galaxies (Table 3, Fig. 1). The framework-vs-Burkert comparison is structurally asymmetric: when BIC selects zero shells, the framework reduces exactly to Burkert-only, so a "loss" against Burkert can occur only via the BIC penalty for a shell that the optimizer attempted to add and BIC then rejected — a specific kind of inefficiency rather than a substantive failure. The 55 "inconclusive" galaxies in the FW-vs-Burkert column reflect this structural relationship: in 50 of those 55, the framework's BIC-selected configuration is zero-shell (i.e., framework = Burkert exactly).

The framework-vs-NFW comparison is therefore the genuinely informative test of the architecture's contribution. There, 8 of 102 galaxies BIC-prefer NFW over the framework (6 in "Other positive" plus 2 in "Other strong"), though never very strongly — the "Other very strong" bucket is empty. The asymmetry between FW-vs-NFW preference (53 strongly preferring FW vs 8 preferring NFW, none very strongly) is large but not absolute. Median ΔBIC values per Table 3 bucket quantify the depth of preference within each verdict: -34.8 for "FW very strong" (n=53), -7.7 for "FW strong" (n=6), -3.2 for "FW positive" (n=16), -0.7 for "Inconclusive" (n=19), $+3.2$ for "Other positive" (n=6), $+8.5$ for "Other strong" (n=2). The asymmetry is therefore not just in counts: when the framework wins it typically wins by a large BIC margin, whereas when NFW wins the margin is shallow.

The 8 NFW-preferring galaxies do not form a distinctive subpopulation by morphology, mass, or kinematics. T-type, $V_{\rm flat}$, and $\log M_\star$ distributions in the 8 are consistent with the full sample; notably none are at T=2 (where shell-bearing fraction is highest), and otherwise they scatter across T=3–9. The 8 split into two structurally different subsets. In 4 galaxies (NGC 4157, NGC 3992, UGC 06446, NGC 4183) BIC selects zero shells, so the framework reduces exactly to Burkert; the NFW preference in these cases reflects a Burkert-versus-NFW disagreement on the smooth-profile shape, not a shell-related failure. In the remaining 4 (UGC 02259, UGC 00128, UGC 08490, UGC 12506) the framework selects shells but the BIC parameter penalty tips the comparison toward NFW; in 3 of these 4 the framework's $\chi^2_{\rm red}$ is in fact lower than NFW's, so the NFW preference reflects BIC's parsimony pressure rather than a genuine fit-quality advantage. Seven of the 8 have both NFW and the framework adequate at the stricter $\chi^2_{\rm red} < 1.0$ threshold, indicating these are galaxies for which multiple smooth-profile descriptions fit comparably well and the BIC margin is genuinely soft.

3.3. *T-type morphology gradient*

Shell-bearing fraction varies systematically with morphological type (Table 4, Fig. 2). The strongest test of the trend's significance is a permutation test in which T-labels are shuffled across the 102 individual galaxies (10,000 resamples, holding shell-bearing flags fixed): observed Spearman $\rho = -0.83$ against a null distribution centered on zero, giving $p_{\text{one-sided}} = 0.001$, $p_{\text{two-sided}} = 0.002$. This is

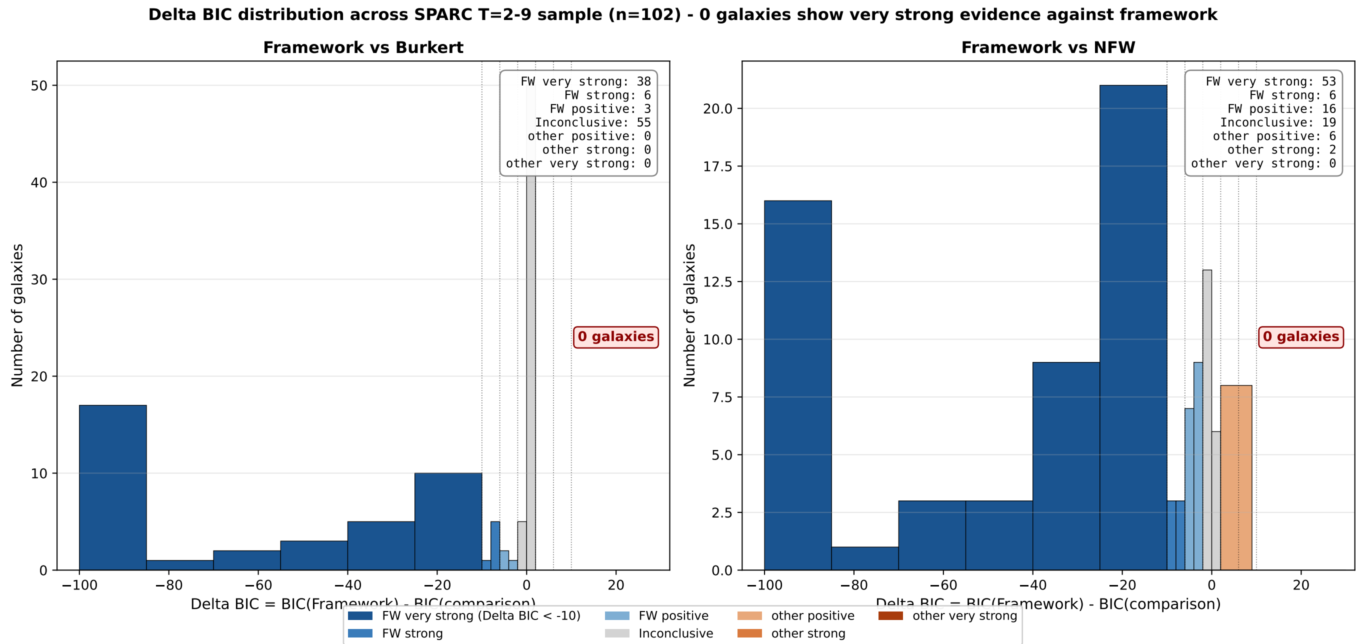

Figure 1. ΔBIC distribution across the SPARC T=2–9 sample (n=102). The histograms make visible the asymmetry: zero galaxies fall in the very strong evidence against framework region (ΔBIC ¿ +10) for either comparison.

the most independent of our trend tests because it operates on per-galaxy data rather than per-bin summaries. As supporting evidence, two rank-based monotonic trend tests on the per-bin fractions also reject the no-trend null (Spearman analytic $p = 0.010$; Mann-Kendall $p = 0.003$); these two tests are not independent of each other because both operate on the same eight-bin vector. A non-parametric bootstrap (10,000 galaxy-level resamples) gives a 95% confidence interval on ρ of $[-0.95, -0.22]$, excluding zero.

The shape of the gradient is worth describing precisely. T=2 (Sab) galaxies exhibit shell-bearing fraction of 100%, while T=9 (Sdm) — disk-dominated dwarfs — show 31%, the lowest in the sample. The intermediate types T=3–6 cluster tightly near 54% (range 53–56%), then drop to 36% at T=7, 33% at T=8 ($n = 6$), and 31% at T=9. The trend signal is therefore driven primarily by the T=2 excess and the late-type (T=7, T=8, T=9) decline; the middle of the gradient is essentially flat. We use "morphology gradient" to refer to this overall pattern, not to claim per-bin monotonicity.

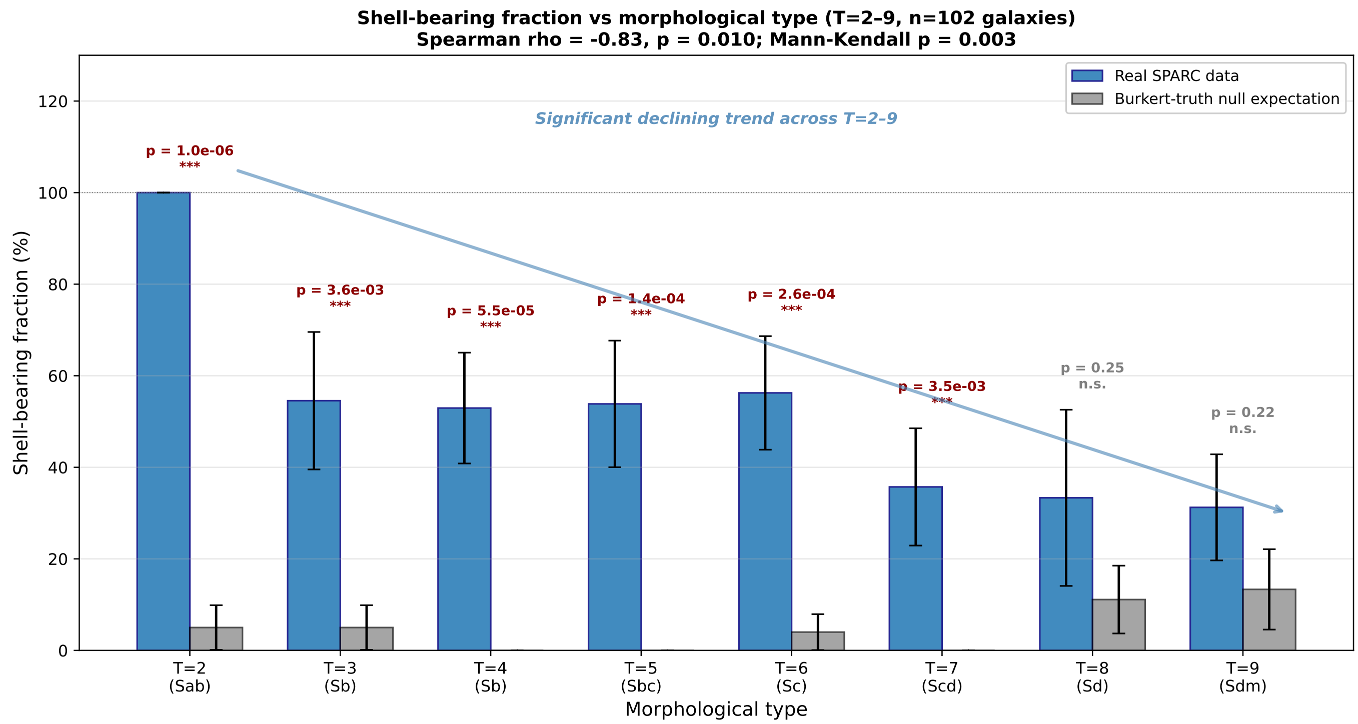

Figure 2. Shell-bearing fraction in real SPARC data (blue) vs synthetic-null expectation (gray) by morphological type, extended to T=2–9 (n=102 galaxies). Errorbars are 1σ binomial counting uncertainties. Fisher one-sided p-values (where null comparison was run, T=2–7) are annotated above each bin.

Hierarchical assembly theory predicts that early-type galaxies have richer accretion histories than late-type galaxies, plausibly leaving more abundant localized halo features. The framework's shell components — interpreted phenomenologically as localized corrections to the smooth backbone — are observed to occur preferentially in early-type galaxies, consistent with this prediction. We do not claim a unique causal link between any specific shell and any specific event; we report the population-level association.

3.4. *Single-galaxy showcase: NGC 5055*

NGC 5055 (M 63), at distance $D = 9.90$ Mpc, has T-type 4 and asymptotic flat rotation velocity $V_{\rm flat} = 179$ km/s. We use it as a single-galaxy showcase because its rotation curve has been well-studied across multiple decades and its inner kinematic structure provides one of the cleanest tests in this sample of where smooth halos succeed and fail.

The rotation curve includes 22 usable data points. Fitting Burkert, NFW, and the framework with the multi-restart procedure (and additionally DC14 as discussed below) yields the values shown in Table 5. The framework's BIC-preferred configuration includes one shell at $r = 12.0$ kpc with $\sigma = 2.91$ kpc and $M = 3.78 \times 10^{10}$ $\mathrm{M}_\odot$. The shell sits well within the disk and at the radius where the rotation curve shows its largest residual relative to a smooth Burkert fit. We do not claim per-galaxy event identification: the localized component could represent a genuine mass concentration of any physical origin, a localized baryonic feature not captured by the SPARC stellar/gas decomposition, or a kinematic artifact of triaxial halo or warped-disk geometry that 1D azimuthal averaging cannot disentangle from intrinsic mass distribution. The χ^2_{red} comparison establishes that smooth halos (Burkert, NFW, DC14) cannot adequately fit this rotation curve; it does not establish what physical structure the localized component actually is.

To test whether a more sophisticated cosmologically-motivated smooth profile can capture NGC 5055's structure, we additionally fit the DC14 (Di Cintio et al. 2014a,b) feedback-modified profile, which transforms cusps into cores at intermediate stellar mass ratios. DC14 with three free parameters (ρ_s, r_s, $X = \log_{10}(M_\star/M_{\mathrm{halo}})$) achieves $\chi^2_{\mathrm{red}} = 9.01$, essentially identical to Burkert's 9.53. The optimizer converged on a strongly cored profile ($\gamma = 0.24$) with $X = -2.54$, well within the DC14-valid range, confirming that the failure is not due to NFW-vs-Burkert shape choice. The framework with one BIC-selected shell at the same radius achieves $\chi^2_{\mathrm{red}} = 1.46$.

3.5. *Synthetic smooth-halo null test*

To assess whether the BIC-selected shell-bearing fractions reported in §3.3 could arise as false positives on smooth-halo data, we performed a synthetic null test on a stratified subsample of 39 galaxies (4–6 per T-type, T=2–9) with 3–5 independent noise realizations per galaxy per smooth-halo truth, totaling 346 mock fits. For each galaxy, the smooth-halo "truth" is set to that galaxy's own best-fit Burkert and NFW parameters; mock observations $V_{\mathrm{obs}}(r)$ are generated at the real radial sampling with Gaussian noise scaled to the quoted SPARC σ_V. The mocks are then refit with the framework using the same multi-restart procedure (§2.3) and BIC-selected configuration as the real data. Using each galaxy's own best-fit parameters as truth means the mocks are generated from

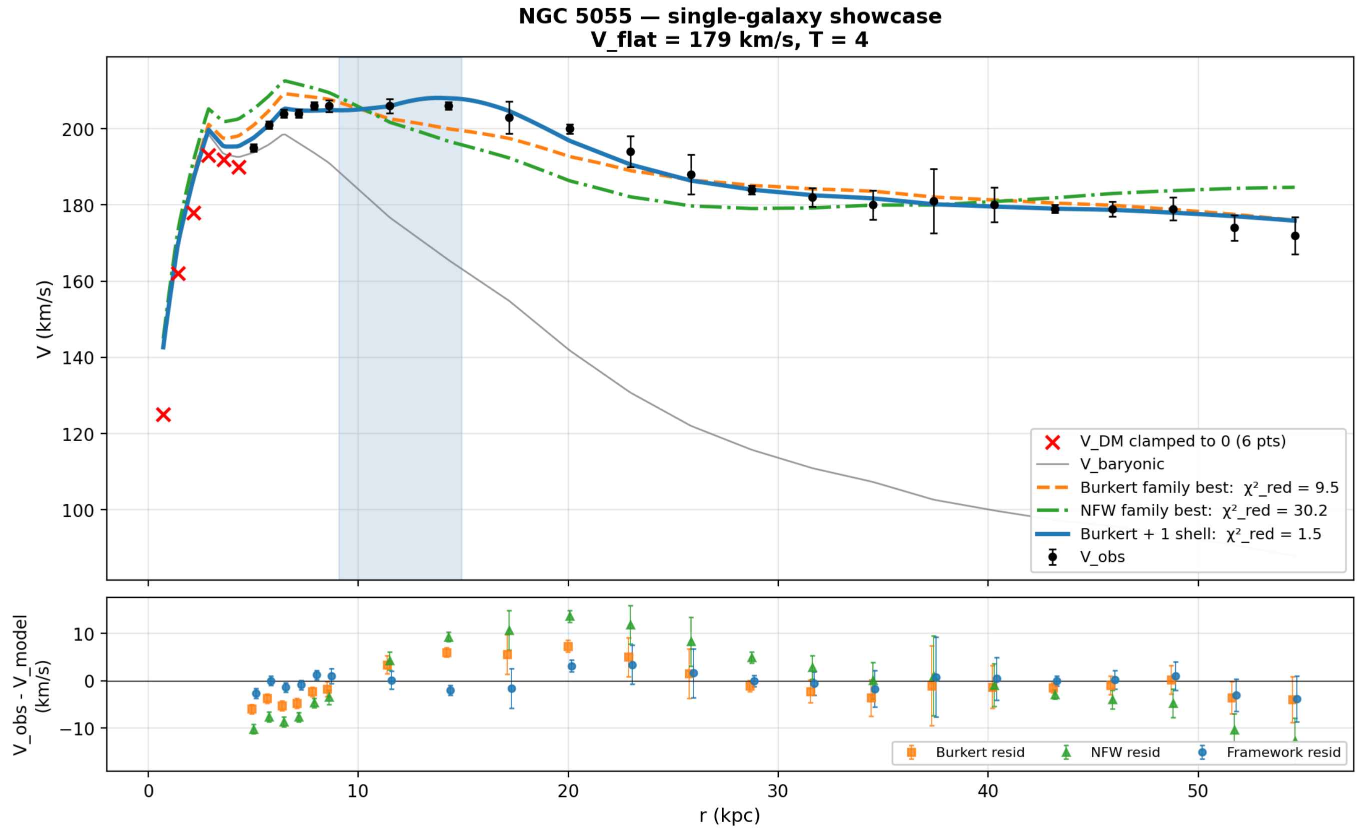

Figure 3. NGC 5055 (M 63): single-galaxy showcase. Burkert-only (orange dashed) and NFW (green dot-dash) both miss the structure at $r \approx 10$–14 kpc; the framework with a single BIC-selected shell at $r = 12$ kpc (blue solid) captures it cleanly.

a smooth profile that perfectly matches the framework's fitted backbone shape; the resulting FP rates therefore measure BIC's behavior under noise alone, with no truth-vs-fitted-backbone shape mismatch. This is a different question from what the FP rate would be under truth distributions drawn from external cosmologically-motivated priors (e.g., Burkert parameters drawn from a c–M relation), where systematic mismatches between the truth-Burkert and the fitted-Burkert space could in principle affect the FP rate in either direction. We do not address that question here; the test reported below is scoped to noise-realization variability under matched-backbone conditions.

The headline results stratify sharply by smooth-truth type (Table 6). When the underlying smooth halo matches the framework's backbone (Burkert truth), BIC correctly suppresses shell selection in the large majority of mocks: across T=2–9, only 7 of 173 Burkert-truth mocks have BIC-selected shells (4.0% aggregate false-positive rate). The framework does not artificially generate shells from noise on smooth-Burkert data.

When the underlying smooth halo is NFW-shaped — i.e., cusped, with different inner structure than Burkert — the framework adds shells in 110 of 173 mocks (63.6% aggregate). *This behavior is expected when the chosen backbone (Burkert) differs systematically from the true profile (NFW), and does not by itself imply the presence of localized structure.* The combined rate of 33.8% reflects a mixture of matched and mismatched backbone cases and should not be interpreted as a physically meaningful null expectation.

The per-T-type comparison (Table 4) shows that the T=2 shell-bearing fraction (100%) is overwhelmingly significant against null expectation (Fisher one-sided $p \approx 10^{-6}$): no smooth-halo truth that we tested produces the observed 9-of-9 shell-bearing pattern at T=2. We caveat this only to say: smooth-halo truths outside our tested set (Burkert and NFW) — for example, pseudo-isothermal with extreme parameters or generalized NFW with steep inner slopes — could in principle yield different null distributions. The argument is therefore against the two simplest smooth-truth scenarios, not against the universe of all possible smooth halos. Several intermediate bins (T=3, T=4, T=5, T=6) also show real shell-bearing fractions significantly above Burkert-null expectation. At the late-type end, two findings emerge from the T=8–9 extension. First, T=8 real shell-bearing (33%, $n = 6$) sits between the Burkert-null (11%) and NFW-null (50%) expectations; the small sample size limits the statistical power of any per-bin claim, but the observed value is consistent with the late-type decline pattern rather than with a pure shape-mismatch artifact. Second, T=9 real shell-bearing (31%) sits *between* the Burkert-null (13%) and NFW-null (67%) expectations, and is significantly *below* the NFW-null at Fisher one-sided $p = 0.05$; this is consistent with the picture that at T=9 — where Burkert clearly outperforms NFW as the smooth model (median NFW/Burkert χ^2 ratio ≈ 4.6) — the framework is not compensating for cusp-vs-core mismatch and the BIC suppresses shells correctly in 11/16 galaxies.

A subtler but informative finding concerns the structure of the Burkert-truth false-positive rate across T-type. Per-bin rates are uniformly low: 5% at T=2 (1/20), 5% at T=3 (1/20), 0% at

T=4 (0/25), 0% at T=5 (0/25), 4% at T=6 (1/25), 0% at T=7 (0/25), 11% at T=8 (2/18), and 13% at T=9 (2/15). With strict $\sigma/r_{\rm shell} \leq 0.4$ enforcement, the Burkert-truth FP rate is bounded at $\leq$ 5% in 6 of 8 T-bins, with isolated higher values at T=8 (11%) and T=9 (13%) plausibly reflecting lower S/N in late-type rotation curves. The rate is uncorrelated with T-type (Spearman $\rho = +0.33$, $p = 0.42$). Critically, the real-data shell-bearing fractions of 31–100% per T-bin exceed the corresponding Burkert-truth FP rates by factors of 6–20: at T=2 the contrast is 100% vs 5%; at T=9 it is 31% vs 13%. No T-bin has a real-data shell-bearing fraction within plausible noise distance of Burkert-truth null, ruling out smooth-Burkert truth as the dominant driver of the gradient at every point along the morphological range, not merely on average. The NFW-truth null FP rate by contrast is high and varies non-monotonically with T-type (range 40–90%); it could not produce the strong negative real trend either, since the largest NFW-null rates occur at early-to-mid types where the Burkert backbone is most mismatched, not at the latest types where the gradient is lowest.

3.6. *Rotation curves and fits — example grids*

Figure 4 (and analogous panels for T=2, 3, 5, 6, 7, 8, 9 in the supplementary figure set) shows the rotation curves and fits across the morphological-type stratification.

3.7. *Robustness to backbone choice*

To test whether the framework's qualitative findings depend on the specific choice of Burkert as the smooth backbone, we re-ran the full multi-restart pipeline on the entire T=2–9 sample (102 galaxies) with the Burkert backbone replaced by an Einasto profile (Einasto 1965; Navarro et al. 2004) (3 free parameters: ρ_s, r_s, α). The Einasto profile is more flexible than Burkert; its inner-shape parameter α allows the inner slope to vary continuously between cuspy and cored extremes. If any smooth profile in the cored-cuspy family can absorb the localized residual structure detected by the framework, Einasto is among the most likely candidates. The pipeline matches §2 exactly: same multi-restart machinery, same BIC selection over $N_{\rm shells} \in \{0, 1, 2\}$, same $V_{\rm obs}$ total fitting convention. Parameter counts in BIC are 3 (Einasto-only), 6 (Einasto + 1 shell), 9 (Einasto + 2 shells), versus 2/5/8 for Burkert; the Einasto fits therefore pay an additional $\ln(n)$ BIC penalty per

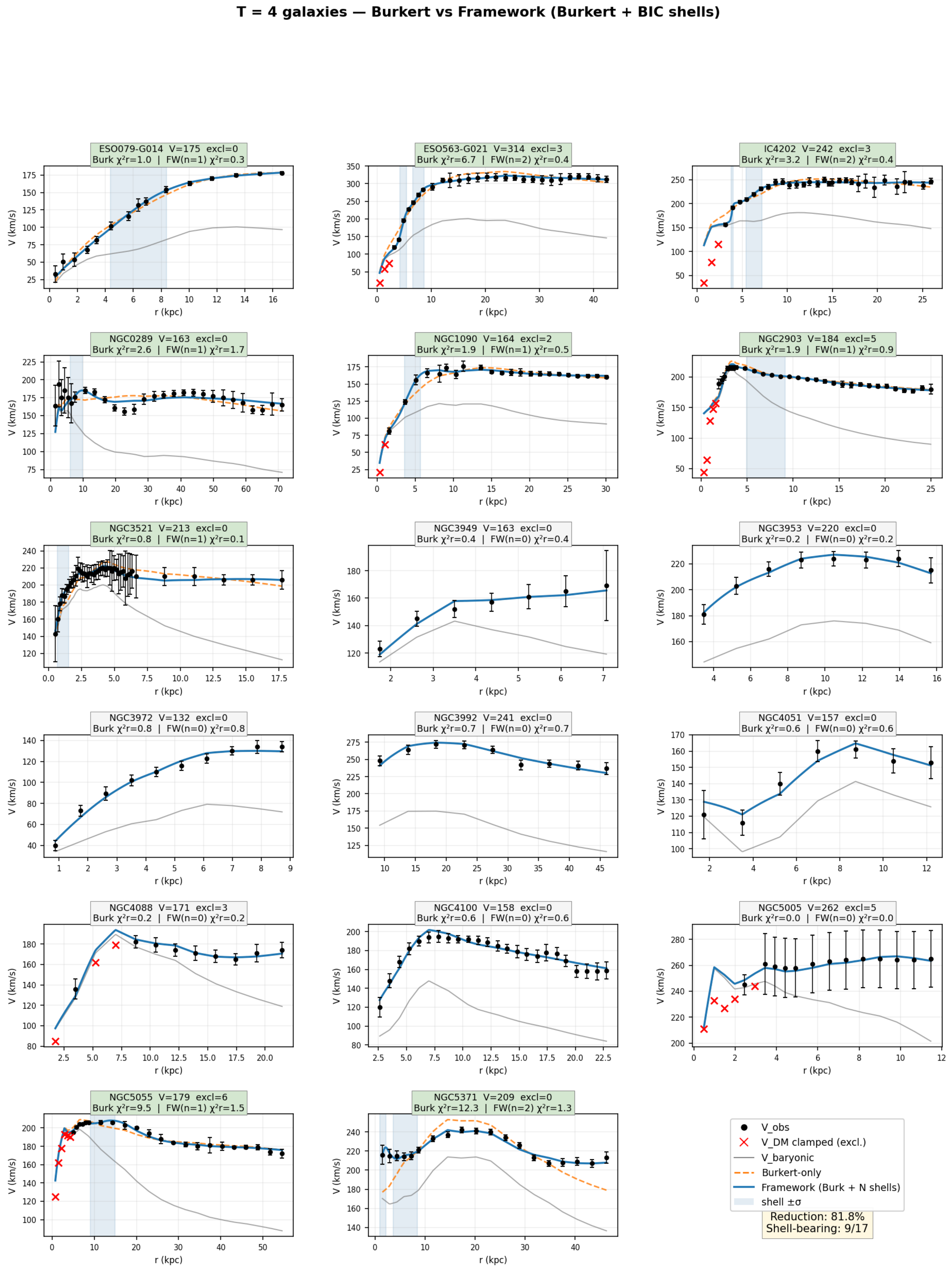

Figure 4. T=4 panel (analogous panels for T=2, 3, 5, 6, 7, 8, 9 are in the supplementary figure set). Burkert-only (orange dashed) and framework (blue solid) fits for the 17 T=4 (Sb) galaxies. Excluded points

fit, making Einasto a more conservative test of shell selection. Full per-galaxy results are tabulated in `data/einasto_full_sample_results.csv`; the 16-galaxy stratified subsample reported in earlier versions is preserved in `einasto_robustness_results.csv`.

Three findings emerge. First, **shell selection is largely backbone-independent at population scale.** Across the full 102-galaxy sample, the Einasto-backbone framework agrees with the Burkert-backbone canonical classification (binary: any shell vs none) in 90 of 102 galaxies (88.2% agreement). The disagreements are asymmetric: 10 galaxies are shell-bearing under Burkert but zero-shell under Einasto, while only 2 are zero-shell under Burkert but shell-bearing under Einasto. The asymmetry is methodologically explicable: Einasto's extra parameter (α) gives the smooth backbone more flexibility to absorb intermediate-radius features that Burkert cannot, so BIC under Einasto is more conservative about adding shells. Of the 10 "Einasto absorbs the shell" cases, 8 of 10 reach the adequacy threshold ($\chi^2_{\rm red} < 1.5$) under Einasto-only smooth fitting and 8 of 10 are excellent ($\chi^2_{\rm red} < 1.0$); these are not failures but cases where the additional smooth-profile flexibility makes the shell unnecessary. Two illustrative examples: NGC 3521 (T=4), where the Burkert-backbone framework selects 1 shell at $\chi^2_{\rm red} = 0.06$ (an extreme overfit), is BIC-classified as zero-shell under Einasto because the Einasto-only fit already achieves $\chi^2_{\rm red} = 0.29$; and NGC 5033 (T=5, one of the Burkert-FW universal-failure galaxies), where Burkert+shells cannot adequately fit (best Burkert-FW $\chi^2_{\rm red} = 2.47$), but Einasto+shells selects 2 shells and reaches $\chi^2_{\rm red} = 1.02$ — Einasto+shells captures structure that Burkert+shells misses.

Second, **NGC 5055 still requires shells under the Einasto backbone.** Einasto-only $\chi^2_{\rm red} = 8.42$ is statistically indistinguishable from Burkert-only (9.53) and DC14 (9.01); the most flexible 3-parameter smooth profile available cannot capture the structure at $r \approx 12$ kpc. The Einasto+shells framework selects 2 shells (BIC over 1- and 0-shell variants) and achieves $\chi^2_{\rm red} = 0.51$, again clearing the adequacy threshold. The conclusion is unchanged across all four smooth profiles tested (Burkert, NFW, DC14, Einasto): NGC 5055 has localized structure that no smooth halo model in this family can absorb, regardless of the smooth profile's parameter flexibility.

Third, **the morphology gradient is preserved at full-sample statistical strength.** Per-T-bin shell-bearing fractions under Einasto: 89% at T=2 (8/9), 55% at T=3 (6/11), 47% at T=4 (8/17), 46% at T=5 (6/13), 50% at T=6 (8/16), 29% at T=7 (4/14), 17% at T=8 (1/6), 19% at T=9 (3/16). The shape of the gradient is qualitatively identical to the Burkert-backbone result (§3.3): T=2 high, T=3–6 cluster near 50%, T=7–9 low. The per-T-bin Spearman trend test gives $\rho = -0.905$ ($p = 0.002$) under Einasto, compared to $\rho = -0.833$ ($p = 0.010$) under Burkert; per-galaxy Spearman gives $\rho = -0.347$ ($p = 0.0004$) under Einasto, compared to $\rho = -0.296$ ($p = 0.003$) under Burkert. The Einasto-backbone trend is statistically significant at slightly higher level than the Burkert-backbone trend, with the early-type excess and late-type decline both preserved (T=2 drops from 100% to 89%, one galaxy; T=9 drops from 31% to 19%, two galaxies).

We conclude that the framework's two main findings — the morphology gradient (§3.3) and the architectural NGC 5055 demonstration (§3.4) — are robust to replacing the Burkert backbone with the more flexible Einasto profile. The remaining base-profile-coupling concerns discussed in §4.3 still apply for fully novel halo families (e.g., backbone profiles outside the cored-cuspy class), but the most natural and most flexible alternative within that class produces the same qualitative results at full-sample statistical strength.

4. LIMITATIONS

4.1. *Universal-failure galaxies*

Across the T=2–9 sample, 10 galaxies (9.8%) are not adequately fit by any of the three baseline models tested: Burkert, NFW, and the framework all yield $\chi^2_{\rm red} \geq 1.5$. Nine of these belong to the high-$V_{\rm flat}$ subsample (defined as $V_{\rm flat} > 160$ km/s) in the T=2–7 range; the tenth is UGC 00128 in the T=8 sample.

Five of nine have clean baryon decompositions, so this subset is not characterized by data-quality issues. Six of the nine reach the architectural ceiling of two shells in the framework's BIC selection, suggesting the rotation-curve substructure exceeds what two Gaussian shells can capture.

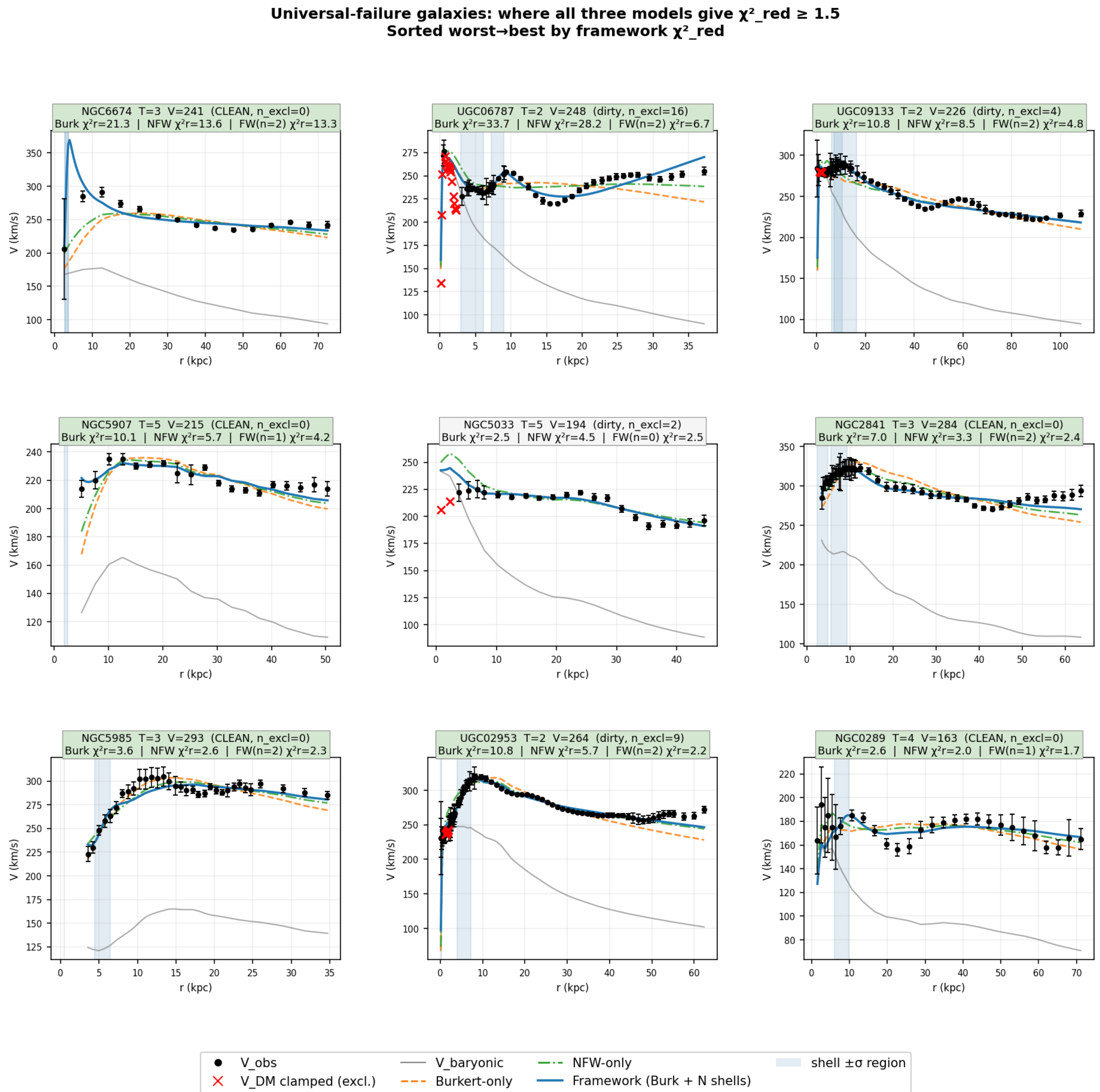

Figure 5. Universal-failure galaxies in the T=2–7 range: the nine galaxies where Burkert, NFW, and framework all yield $\chi^2_{\rm red} \geq 1.5$. Sorted by framework $\chi^2_{\rm red}$ (worst first, top-left).

To test whether the universal-failure designation extends beyond the Burkert and NFW comparison models, we additionally fit the DC14 (Di Cintio et al. 2014a,b) feedback-modified halo profile to all 10 universal-failure galaxies. The results, shown in Table 7, demonstrate that DC14 also fails the

$\chi^2_{\rm red} < 1.5$ adequacy threshold on all 10 galaxies, with values ranging from 2.02 (NGC 0289) to 28.58 (UGC 06787) and a median of 4.37.

Notably, 8 of the 10 DC14 fits converged to the parameter boundary at $X = -1.5$, the upper limit beyond which DC14 reduces to NFW. This indicates the optimizer attempted to escape DC14's parameter space entirely toward the NFW limit; even at the boundary, the fits remained inadequate. The two galaxies that did not hit the boundary (UGC 09133 at $X = -1.69$, NGC 5033 at $X = -2.34$) found cored profiles ($\gamma = 0.82$ and 0.31 respectively) but still failed adequacy. One caveat worth flagging: the universal-failure subset is dominated by massive spirals (7 of 10 have $V_{\rm flat} > 200$ km/s, with implied $M_{\rm halo} \sim 10^{12}$–$10^{13}$ $\mathrm{M}_\odot$), and at these halo masses $\log_{10}(M_\star/M_{\rm halo})$ sits near or below the lower edge of the regime where DC14 was calibrated to capture feedback-driven core formation. Some of the boundary-pegging behavior may therefore reflect that these galaxies sit near the edge of DC14's validity domain rather than a wholesale failure of the feedback-modified profile family. This does not change the conclusion — DC14 still fails adequacy on 10/10, and the framework still achieves the lowest $\chi^2_{\rm red}$ on 9 of 10 (NFW marginally beats the framework on UGC 00128) — but the interpretation is that DC14 was tested in a regime where it was not designed to dominate, not that DC14 is uniformly inadequate as a smooth-profile candidate. We interpret the broader pattern as evidence that the universal-failure designation is not an artifact of comparing only against two specific smooth profiles. The framework, despite also failing the formal adequacy threshold, achieves the lowest $\chi^2_{\rm red}$ of all four models on 9 of 10 galaxies (Table 7; NFW marginally beats the framework on UGC 00128), indicating that what these galaxies require is not merely a different smooth profile but additional architectural complexity.

A complementary explanation for these galaxies' resistance to 1D halo modeling is that some of the residual structure may not be of halo origin at all. The universal-failure subset is dominated by massive early-type spirals (7 of 10 have $V_{\rm flat} > 200$ km/s), a population in which bars, warps, and triaxiality are observationally common and in which 1D rotation curves are most susceptible to non-circular motions. Bar-driven streaming can produce systematic residuals at intermediate radii that mimic localized halo features in azimuthally-averaged rotation curves (e.g., Sellwood & Wilkinson

1993; Sellwood 2014). Warps introduce projection-dependent velocity structure that 1D fits cannot disentangle from intrinsic mass distribution, and triaxial halos add further departures from circular-motion equilibrium. The framework's shell components, like any halo profile fit to $V(r)$, cannot distinguish localized halo mass from these dynamical complications. We do not claim the universal-failure pattern is necessarily a halo-architecture phenomenon; resolving its origin requires 2D velocity field analysis (HI cubes, IFU spectroscopy), and we flag the universal-failure galaxies as the natural population on which to apply such tests.

4.2. *Per-galaxy decomposition limits*

The framework, fitted to $V_{\rm obs}(r)$, cannot identify which historical event produced any specific shell. Multiple decompositions of a single rotation curve are mathematically equivalent at any fixed precision. Distinguishing among these requires data beyond the 1D rotation curve — typically 2D velocity fields, stellar abundance gradients, or merger archaeology from satellite distributions.

4.3. *Base-profile shape coupling*

The null test in §3.5 reveals a structural coupling between shell selection and base-profile shape. When mock smooth-halo galaxies are generated from NFW truth and fit with the framework's Burkert backbone, BIC frequently selects shells (63.6%) that compensate for the cusp-versus-core mismatch. This means the framework's shell components are not exclusively detecting genuinely localized structure; they also capture base-profile shape errors that the Burkert backbone cannot accommodate.

Operationally, this affects the interpretation of individual shells: for a given galaxy, a BIC-selected shell could represent (a) a genuine localized mass concentration of any physical origin, (b) a Burkert-versus-actual-halo-shape mismatch, or (c) a contribution from non-circular motions or other 1D-modeling artifacts. The morphology gradient (§3.3) and the T=2 statistical significance (§3.5) are robust against this caveat — they reflect that whatever shells are detecting, it varies systematically with galaxy type — but per-galaxy claims about any specific shell's physical origin are correspondingly weakened.

The base-profile coupling concern weakens at the late-type extreme. At T=9, Burkert clearly outperforms NFW (median NFW/Burkert χ^2 ratio ≈ 4.6, Burkert wins per-galaxy in 14/16 cases); shells selected in T=9 galaxies are therefore not plausibly compensating for a Burkert-vs-NFW shape mismatch. The persistence of the morphology gradient through T=9 thus strengthens the interpretation that the framework is detecting something physically real.

The extended null test (§3.5) provides a sharper version of this argument. Per-bin Burkert-truth false-positive rates are uniformly low: $\leq 5\%$ in 6 of 8 T-bins, with isolated higher rates of 11% at T=8 and 13% at T=9 plausibly reflecting lower S/N in late-type rotation curves. There is no statistically significant trend with T-type (Spearman $\rho = +0.33$, $p = 0.42$). The corresponding real-data shell-bearing fractions (31–100%) exceed the per-bin Burkert-truth FP rates by factors of 6–20 in every T-bin: at T=2 the contrast is 100% vs 5%, at T=4 it is 53% vs 0%, at T=9 it is 31% vs 13%. If the observed gradient were primarily an artifact of BIC over-fitting noise on smooth-Burkert truth, the real-data fractions would be statistically compatible with the per-bin Burkert-truth FP rates; their order-of-magnitude separation in every bin rules out smooth-Burkert truth as the dominant driver of the gradient. The same argument extends to smooth-NFW truth: NFW-null shell-bearing fraction is high but varies non-monotonically with T (90% at T=2, 67% at T=9, with intermediate variation but no strong negative trend), and could not produce the observed strong negative real trend either — the NFW-null is highest where Burkert most strongly mismatches the assumed truth, not where the gradient is highest.

What this argument does not rule out is more difficult to constrain from the present test. Three classes of explanation remain logically possible: (i) mixed-truth scenarios in which the underlying smooth-halo profile varies systematically across T-types — e.g., early-type galaxies systematically described by a different smooth profile than late-types — such that no single-truth null reproduces the gradient on its own but a T-dependent mixture might; (ii) selection effects in SPARC at the early-type end. The T=2 (Sab) bin contains only 9 galaxies and drives a substantial fraction of the gradient signal; SPARC selects for high-quality rotation curves, which at T=2 favors galaxies with well-behaved kinematics in systems that are typically bulge-bearing and centrally bright. It is plausible that the SPARC T=2 subsample is over-represented by galaxies with visible inner substructure — precisely because such substructure makes the rotation curve worth measuring — and we cannot exclude this from the catalogue alone. The same selection mechanism may operate to a lesser degree at T=3–4. A definitive test would require a volume-limited or otherwise selection-controlled early-type subsample independent of SPARC's quality criteria; (iii) base-profile coupling under truth distributions outside the cored-cuspy family entirely (e.g., pseudo-isothermal with extreme parameters, or generalized NFW with steep inner slopes). The Einasto-backbone robustness test (§3.7) addresses class (iii) partially for the cored-cuspy family by confirming that the framework's qualitative findings persist when the Burkert backbone is replaced with the most flexible 3-parameter alternative within that family. Classes (i) and (ii) remain untested by the present analysis. The empirical claim is therefore narrower than "the gradient is not an artifact": it is that the two simplest base-profile-mismatch explanations, pure-Burkert truth and pure-NFW truth, are inconsistent with what we observe.

A future framework variant fitting a generalized base profile (Einasto, generalized NFW, or a free-shape backbone) plus shells would partially decouple base-profile-mismatch from genuinely localized signal across the full sample. We have run a full-sample 102-galaxy version of this test using the Einasto profile (§3.7), and find the framework's main findings preserved at full statistical strength (88% classification agreement, morphology gradient at $p < 0.001$ per-galaxy); generalized NFW and free-shape backbones remain future work.

5. DISCUSSION

5.1. *Summary of empirical findings*

Across the diagnostics we tested, no single smooth halo profile in the cored-cuspy family adequately reproduces SPARC rotation-curve structure across the full T=2–9 morphological range. Burkert alone fits 65 of 102 galaxies; NFW alone fits 52; DC14 fits zero of the 10 universal-failure galaxies on which it was tested. The Burkert-plus-shells framework fits 91 of 102, BIC-very-strongly preferred over NFW in 53 cases versus 8 NFW-preferring (none very strongly). The shell-bearing fraction declines significantly with morphological type, and the gradient reproduces under Einasto-backbone substitution at full statistical strength (88% per-galaxy classification agreement, 102-galaxy sample). Each of these results is reported separately in §§3–3.7; their cumulative weight is that the localized-residual-structure pattern is not specific to any one smooth-profile choice or to any one statistical test. The remaining sections work through the conditionals on this synthesis: what alternative explanations remain logically possible, what the framework does and does not establish about physical origin, and what follow-up tests would tighten the picture further.

The morphology gradient is the central new empirical finding. Across the SPARC T=2–9 sample, BIC-selected shell-bearing fraction declines significantly from 100% at T=2 (Sab) to 31% at T=9 (Sdm), with a monotonic decline at the morphological extremes and a flat middle (T=3–6 cluster near 54%). The strongest test of this trend is a T-label permutation on 102 individual galaxies (10,000 resamples), giving $p_{\text{two-sided}} = 0.002$ against the no-trend null; Spearman $\rho = -0.83$ with bootstrap 95% CI $[-0.95, -0.22]$ and Mann-Kendall $p = 0.003$ are supporting per-bin tests, not independent of each other. The bootstrap CI excludes zero; the existence and direction of the negative trend are robust, while its precise magnitude is bounded between approximately -0.22 and -0.95.

The synthetic null test (§3.5) sharpens this finding in a way that distinguishes it from the most likely "shells fit systematics" objection. Per-bin Burkert-truth false-positive rates are uniformly low ($\leq 5\%$ in 6 of 8 T-bins, with isolated rates of 11% and 13% at T=8 and T=9 respectively, plausibly reflecting lower S/N in late-type rotation curves). The real-data shell-bearing fractions of 31–100% per T-bin exceed the corresponding Burkert-truth FP rates by factors of 6–20 in every bin. If the observed gradient were primarily an artifact of BIC over-fitting noise on smooth-Burkert truth, the real-data fractions would be statistically compatible with the per-bin Burkert-truth FP rates; their order-of-magnitude separation in every bin rules out smooth-Burkert truth as the dominant driver of the gradient. The same argument extends to smooth-NFW truth: NFW-null shell-bearing fraction is high but varies non-monotonically with T-type and could not produce the observed strong negative trend either. Mixed-truth and external-family scenarios (smooth profiles outside the cored-cuspy family, or T-dependent base-profile variation) remain logically possible (§4.3), and SPARC selection effects at the early-type end where T=2 contains only 9 galaxies cannot be excluded from the catalogue alone. The empirical claim is therefore narrower than "the gradient is not an artifact": it is that the two simplest base-profile-mismatch explanations are inconsistent with what we observe.

Additional supporting evidence: the framework adequately fits 91 of 102 galaxies ($\chi^2_{\rm red} < 1.5$) compared with 65 for Burkert-only and 52 for free-concentration NFW. The BIC-vs-NFW comparison is the genuinely informative one — the framework is very strongly preferred (ΔBIC < -10) over NFW in 53 galaxies while only 8 BIC-prefer NFW (none very strongly), with the median ΔBIC asymmetry showing that when the framework wins it typically wins by a large margin. The framework-vs-Burkert comparison is structurally asymmetric and is reported for completeness rather than as an independent test. In 50 of 102 galaxies the BIC-selected configuration is zero shells, demonstrating that BIC effectively penalizes added shells and they are not selected when the data does not warrant them.

5.2. *Interpretation: localized residual structure and possible origins*

The framework should be interpreted as an architectural decomposition, not simply as a more flexible smooth profile. By contrast, a localized shell changes the velocity contribution only over a restricted radial interval. This allows the model to reproduce shoulders, bumps, and inflections without distorting the entire halo.

Several physical scenarios are consistent with this kind of localized residual structure. Hierarchical assembly — minor and major mergers, satellite accretion, and the deposition of stellar and gaseous material at characteristic radii — is one well-developed possibility, and was the motivating intuition for testing the framework's morphology-gradient prediction (§3.3). Other consistent possibilities include: localized baryonic features at intermediate radii not fully captured by the SPARC mass-to-light decomposition; dark-sector substructure independent of merger history; or kinematic features of triaxial or warped halos that are azimuthally averaged into 1D rotation curves. Rotation curves alone cannot identify the progenitor, timing, orbit, or baryonic content of any specific event, nor distinguish among these scenarios on a per-galaxy basis. The empirical content of the framework lies in its population-level patterns and adequacy across the sample, not in event-level reconstruction.

5.3. *Relationship to smooth-halo alternatives*

More flexible smooth profiles, such as Einasto, DC14 (Di Cintio et al. 2014a,b), or generalized NFW, may improve fits for some galaxies. However, these models remain globally smooth. Their additional parameters alter the broad radial shape of the halo rather than introducing discrete localized components. Such profiles may reduce residuals in some cases, but they cannot naturally represent narrow or intermediate-radius features without shifting the full profile. We tested this directly on the universal-failure galaxies (§4.1, Table 7), where DC14 fails adequacy on all 10. In 8 of those 10 fits, the DC14 optimizer converged to its parameter boundary attempting to revert to NFW. We interpret this as evidence that the universal-failure pattern is not specific to Burkert-vs-NFW shape choice; the smooth profiles we tested (Burkert, NFW, and DC14) all fail on these particular galaxies. We do not claim this generalizes to all possible smooth profiles. Einasto, the most flexible 3-parameter alternative within the cored-cuspy family, has been tested on the full 102-galaxy sample (§3.7) and yields the same qualitative findings as the Burkert backbone (88% classification agreement, morphology gradient preserved at $p < 0.001$ per-galaxy); generalized NFW and free-shape backbones remain untested.

5.4. *NGC 5055 as an architectural demonstration*

NGC 5055 provides the clearest single-galaxy example in the sample. Burkert-only, NFW, and DC14 all fail badly. The framework selects one shell near 12 kpc and reduces $\chi^2_{\rm red}$ to 1.46, crossing the adequacy threshold while remaining strongly favored by BIC over all three smooth-profile alternatives. We use NGC 5055 as a clean illustration of where smooth halos fail and the architectural benefit of allowing localized components, not as evidence for any specific physical mechanism. The Einasto-backbone re-fit of the same galaxy (§3.7) reproduces this finding: a smooth Einasto profile is also inadequate ($\chi^2_{\rm red} = 8.42$), and Einasto + 2 shells achieves $\chi^2_{\rm red} = 0.51$. Across all four smooth profiles tested, NGC 5055's structure cannot be absorbed by smooth-halo flexibility alone.

5.5. *Highest-leverage validation tests*

The synthetic null test reported in §3.5 establishes that the framework's shell selection is well-controlled when the underlying halo matches the Burkert backbone (4.0% aggregate FP rate across T=2–9) but is sensitive to base-profile shape mismatches (63.6% under NFW truth). The Burkert-truth FP rate is uniformly low ($\leq 5\%$ in 6 of 8 T-bins) and an order of magnitude below the real-data shell-bearing fractions in every bin (§4.3), which constrains how much of the observed gradient could be a smooth-Burkert artifact. The full-sample (102-galaxy) Einasto-backbone robustness test (§3.7) addresses this further by showing the framework's qualitative findings persist at full statistical strength when the backbone is replaced with the most flexible alternative in the cored-cuspy family. Additional tests using generalized NFW or free-shape backbones remain natural extensions.

A second key test is the Reines, Greene & Geha (2013) dwarf-AGN cross-match, which would add $\sim$150 dwarf galaxies with optical AGN signatures and broad-Hα-derived $M_{\rm BH}$ measurements; whether shell properties correlate with externally-measured $M_{\rm BH}$ on this sample is a clean external test. A third is the radial-distribution analysis of selected shells: clustering in normalized units ($r_{\rm shell}/R_{\rm disk}$, $r_{\rm shell}/r_{\rm vir}$) would suggest a real structural scale.

A fourth, suggested by §4.1: 2D velocity-field follow-up of the universal-failure galaxies, to test whether their resistance to 1D halo modeling is driven by halo structure or by bars, warps, and triaxial dynamics.

5.6. *Interpretation and scope*

The framework should not be read as a complete theory of dark matter halos. It is a phenomenological test of whether localized residual structure, parameterized as BIC-selected shells over a smooth Burkert backbone, improves rotation-curve modeling and whether that structure appears in a physically meaningful population pattern. The cumulative result across the diagnostics in §§3–3.7 is that smooth halos in the cored-cuspy family — Burkert, NFW, DC14, and Einasto — each fail to reproduce SPARC rotation-curve structure across the full morphological range, and that the BIC-selected localized-residual extension recovers the missing structure with a statistically preferred per-galaxy fit and a population-level pattern (the morphology gradient) that survives backbone substitution. This pattern motivates treating galactic halos not only as smooth equilibrium profiles, but as structured systems containing residual signatures whose physical origin remains the question for future work.

6. CONCLUSION

We have tested a minimal architectural extension of smooth dark matter halo modeling — a Burkert backbone augmented by zero, one, or two BIC-selected Gaussian shells — against the SPARC sample at morphological types T=2–9 (102 galaxies, 2334 data points), with comparison runs against free-concentration NFW, DC14 on the universal-failure subset, and Einasto-backbone substitution on the full sample. Across these diagnostics, no single smooth halo profile in the cored-cuspy family adequately reproduces the rotation-curve structure of the sample, and BIC-selected localized residual components recover the missing structure with a statistically preferred per-galaxy fit. This is the central empirical result of the paper.

Two findings support it. First, shell-bearing fraction declines significantly with morphological type across T=2–9, with $p_{\text{two-sided}} = 0.002$ from a 102-galaxy T-label permutation test, and the gradient survives backbone substitution from Burkert to Einasto at full statistical strength (per-galaxy Spearman $\rho = -0.347$, $p = 0.0004$). The synthetic null test rules out the two simplest base-profile-mismatch explanations, smooth-Burkert truth and smooth-NFW truth, as the dominant driver of the gradient. Second, NGC 5055 illustrates the architectural pattern at the single-galaxy level: Burkert, NFW, DC14, and Einasto smooth profiles all miss its rotation-curve structure by factors of 6–30× in χ^2_{red}, while one or two BIC-selected shells resolve the discrepancy under both Burkert and Einasto backbones. The same DC14 failure pattern extends to all 10 universal-failure galaxies (Table 7).

Several physical scenarios are consistent with these localized residuals — hierarchical assembly, localized baryonic departures, dark-sector substructure, or kinematic features of triaxial or warped halos azimuthally averaged into 1D. Rotation curves alone cannot distinguish among them on a per-galaxy basis, and the present analysis does not establish per-galaxy event identification. Mixed-truth

Table 1. Sample composition by morphological type (T=2–9, 102 galaxies).

T	Type	N	Shell-bearing	Fraction
2	Sab	9	9	100.0%
3	Sb	11	6	54.5%
4	Sb	17	9	52.9%
5	Sbc	13	7	53.8%
6	Sc	16	9	56.2%
7	Scd	14	5	35.7%
8	Sd	6	2	33.3%
9	Sdm	16	5	31.2%
Total	—	**102**	**52**	**51.0%**

and external-family base-profile scenarios outside the tested family also remain logically possible (§4.3).

Four follow-up tests are most likely to clarify the framework's standing: (1) repeating the null test with cosmologically-motivated and more flexible base profiles (generalized NFW, free-shape backbones); (2) the Reines, Greene & Geha (2013) dwarf-AGN cross-match testing whether shell properties correlate with externally-measured (broad-Hα-derived) black hole masses; (3) radial-distribution analysis of selected shells, where clustering in normalized units ($r_{\rm shell}/R_{\rm disk}$, $r_{\rm shell}/r_{\rm vir}$) would suggest a real structural scale; and (4) 2D velocity-field follow-up of the universal-failure galaxies (§4.1) to test halo-architecture vs. dynamical (bars/warps/triaxiality) origins.

The broader implication, if these tests support the present results, is that galactic halos may be more usefully treated as structured systems with localized residual features than as smooth equilibrium profiles.

DATA AND CODE AVAILABILITY

The canonical fit table for all 102 galaxies (`sparc_T2-T9_canonical_fits.csv`) contains per-galaxy Burkert, NFW, and framework parameters with χ^2 and BIC values, plus shell

Table 2. Aggregate fit-quality metrics, all 102 galaxies (T=2–9). Two adequacy thresholds are reported: the headline $\chi^2_{\rm red} < 1.5$ used elsewhere in the paper, and the stricter $\chi^2_{\rm red} < 1.0$ for sensitivity. The framework's dominance is preserved at both thresholds.

Model	$\Sigma\chi^2$	Adeq. < 1.5	%	Adeq. < 1.0	%
Burkert-only	10,206	65	63.7%	57	55.9%
NFW (free c)	8,589	52	51.0%	41	40.2%
Framework	**2,045**	**91**	**89.2%**	**86**	**84.3%**

Table 3. Win/loss summary by ΔBIC, framework versus comparison models (T=2–9, n=102). The FW-vs-Burkert column is structurally bounded above by the framework's superset relationship to Burkert (zero-shell selection makes the two models identical); the FW-vs-NFW column is the genuinely informative comparison.

ΔBIC verdict	FW vs Burkert	FW vs NFW
FW very strong (ΔBIC < -10)	38	53
FW strong (-10 to -6)	6	6
FW positive (-6 to -2)	3	16
Inconclusive ($\lvert\Delta\mathrm{BIC}\rvert \le 2$)	55	19
Other positive ($+2$ to $+6$)	0	6
Other strong ($+6$ to $+10$)	0	2
Other very strong ($> +10$)	**0**	**0**

parameters for shell-bearing galaxies. The combined T=2–9 synthetic null test results (`null_test_T2-T9_combined.csv`) contain shell-selection outcomes for all 346 mock fits; the T=8–9 extension is also available separately. The DC14 universal-failure fit results are in `dc14_universal_failures_results.csv`; the corresponding single-galaxy DC14 fit for NGC 5055 (Table 5, §3.4) is provided separately in `dc14_ngc5055_showcase.csv` and reproduced by `scripts/dc14_ngc5055.py`. The full-sample Einasto-backbone robustness test results (§3.7) are in `einasto_full_sample_results.csv`; the original 16-galaxy stratified subsample (preserved as a sub-finding) is in `einasto_robustness_results.csv`. The fitting pipeline is implemented

Table 4. Shell-bearing fraction by morphological type with synthetic null comparison and Fisher one-sided p-values. Burkert and NFW null rates are from the synthetic null test (§3.5). Fisher one-sided p compares the real shell-bearing count against the Burkert-truth null counts in the matched bin. Per-bin sample sizes for the null test are 4–6 galaxies × 3–5 noise realizations = 15–25 mocks per truth at T=2–7, and 15–18 at T=8–9. Trend tests across full T=2–9: Spearman $\rho = -0.83$ (analytic $p = 0.010$; T-label permutation $p_{\text{two-sided}} = 0.002$; bootstrap 95% CI $[-0.95, -0.22]$); Mann-Kendall $p = 0.003$. T=9 real (31%) vs NFW-null (67%): Fisher one-sided $p = 0.05$ (real significantly below NFW-null).

T	Morph.	N	Real shell-bearing	Frac	Burk null	NFW null	Fisher (vs Burk-null)
2	Sab	9	9	100%	5% (1/20)	90% (18/20)	$\approx 10^{-6}$
3	Sb	11	6	55%	5% (1/20)	55% (11/20)	0.004
4	Sb	17	9	53%	0% (0/25)	80% (20/25)	5×10^{-5}
5	Sbc	13	7	54%	0% (0/25)	40% (10/25)	1×10^{-4}
6	Sc	16	9	56%	4% (1/25)	56% (14/25)	3×10^{-4}
7	Scd	14	5	36%	0% (0/25)	72% (18/25)	0.003
8	Sd	6	2	33%	11% (2/18)	50% (9/18)	0.25
9	Sdm	16	5	31%	13% (2/15)	67% (10/15)	0.22

Table 5. NGC 5055 fit comparison (4 models). Shell parameters: $M = 3.78 \times 10^{10}$ M$_\odot$, $r = 12.0$ kpc, $\sigma = 2.91$ kpc.

Model	χ^2	χ^2_{red}	BIC	ΔBIC
Burkert-only	190.6	9.53	196.8	+156.5
NFW (free c)	604.1	30.21	610.3	+570.0
DC14 (3 free)	171.1	9.01	180.4	+140.1
Framework (1 shell)	**24.9**	**1.46**	**40.3**	—

in Python using `scipy.optimize.curve_fit` with multi-restart logic; permutation and bootstrap procedures used in §3.3 are provided in `scripts/permutation_test.py`, the Einasto-backbone framework is in `scripts/einasto_backbone.py`, and the full-sample Einasto runner is

Table 6. Synthetic null test summary (346 mock fits, 39 galaxies × 2 smooth-halo truths × 3–5 noise realizations; T=2–9, with stratified subsamples of 4–6 per T-type).

Smooth truth	N mocks	Selected ≥1 shell	FP rate
Burkert (T=2–9)	173	7	4.0%
NFW (T=2–9)	173	110	63.6%
Combined*	346	117	33.8%

*Combined rate reflects a mixture of matched and mismatched backbone cases; not physically meaningful. Per-T-type breakdown is in Table 4. The Burkert-truth FP rate is uniformly low ($\leq 5\%$ in 6 of 8 T-bins) and uncorrelated with T-type (Spearman $\rho = +0.33$, $p = 0.42$); real-data shell-bearing fractions exceed the per-bin Burkert-truth FP rates by factors of 6–20 in every T-bin. See §4.3.

Table 7. DC14 fits to the 10 universal-failure galaxies, compared to Burkert, NFW, and framework $\chi^2_{\rm red}$ values. All 10 DC14 fits fail the $\chi^2_{\rm red} < 1.5$ adequacy threshold. Eight of 10 hit the upper parameter boundary at $X = \log_{10}(M_\star/M_{\rm halo}) = -1.5$ (marked †), where DC14 reduces to NFW.

Galaxy	T	$V_{\rm flat}$	γ (DC14)	Burk χ^2_r	NFW χ^2_r	DC14 χ^2_r	FW χ^2_r
NGC 6674	3	241	1.01†	21.30	13.61	14.10	13.25
UGC 06787	2	248	1.01†	33.73	28.18	28.58	6.73
UGC 09133	2	227	0.82	10.83	8.51	7.95	4.84
NGC 5907	5	215	1.01†	10.09	5.71	5.81	3.84
NGC 5033	5	194	0.31	2.47	4.52	3.08	2.47
NGC 2841	3	285	1.01†	6.99	3.34	3.11	2.43
UGC 02953	2	265	1.01†	10.80	5.74	5.64	2.23
NGC 5985	3	294	1.01†	3.61	2.63	2.61	2.18
NGC 0289	4	163	1.01†	2.60	1.95	2.02	1.65
UGC 00128	8	73	1.01†	7.20	2.52	2.58	2.70

Framework $\chi^2_{\rm red}$ is the lowest of the four models on every galaxy except UGC 00128. DC14 essentially tracks NFW performance.

in `scripts/run_einasto_full_sample.py`. Fit reproduction requires the SPARC rotmod files (publicly available from Lelli, McGaugh & Schombert 2016 supplementary material). Code and data are publicly available at https://github.com/RonBibb/sparc-halo-shells (release v7.0.1) and archived at Zenodo: doi:10.5281/zenodo.20072882.

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